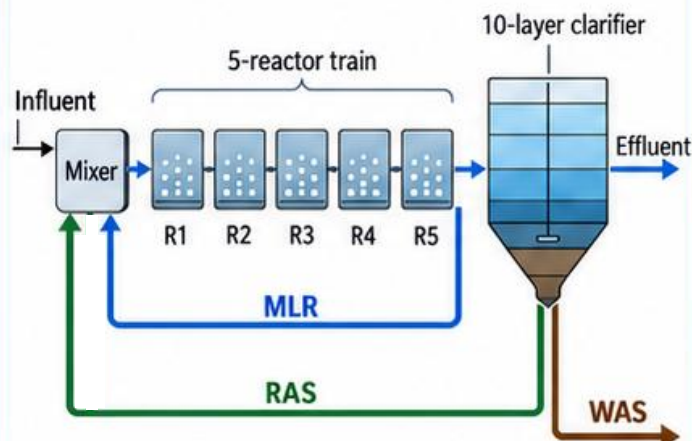


1

Plant Configuration and Operating Controls

Activated sludge process flowsheet



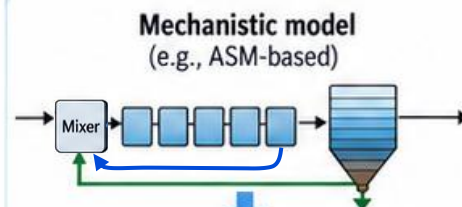
Operating controls (decision variables)

	HRT	Hydraulic retention time (reactor volume)
	Aeration in R3–R5	Air flow rates / DO setpoints in reactors 3–5
	MLR	Mixed-liquor recycle ratio (from R5 to mixer)
	RAS	Return activated sludge ratio (from clarifier underflow to mixer)
	WAS	Waste activated sludge flow (from clarifier underflow)

2

Model Development with Physical Constraints

Generate plant states using mechanistic model



Many plant states across wide operating conditions



Extended ICSOR surrogate

Second-order regression

Joint projection

Enforced physical relations



Connected plant responses projected onto a physically feasible set

3

Two Optimization Pathways

A

Extended ICSOR optimization



- Surrogate-based optimization
- Physically constrained predictions
- Fast computation

B

Smooth NLP

$$\frac{dx}{dt} = f(x, u)$$

- Gradient-based optimization
- Original mechanistic model (smooth)
- Higher computational cost



Independent mechanistic evaluation

Selected operating decisions are checked with the original mechanistic model

Prediction feasibility, prediction accuracy, decision quality, and computation time are assessed separately.

4

Key Findings



Holdout states evaluated:
1,845



Projection reduced aggregate
nRMSE by 8.19%



Projection reduced aggregate
nMAE by 15.65%



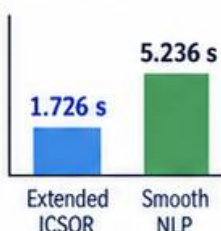
Smooth NLP gave **lower reference-model objective** in the nominal case and all 10 influent scenarios



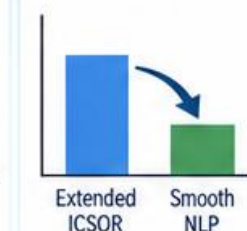
Mean optimization time across 10 scenarios:

Extended ICSOR 1.726 s
Smooth NLP 5.236 s

Computation time (faster is better)



Reference-model objective (lower is better)



Main message: Plant-wide physical projection improves surrogate prediction accuracy and preserves physical consistency, but final operating decisions should still be verified with the mechanistic model.